

Question 1

SELECT DISTINCT department

FROM students;

Department
HR
Finance
IT

Question 2

SELECT department,

AVG (age) **AS** average_age_department

FROM Students

GROUP BY department;

Department	AVE_AGE
HR	22
Finance	22
IT	21

Question 3

SELECT department,

COUNT (*) **AS** student_count

FROM students

GROUP BY department

HAVING BY COUNT(*)>1;

Department	Student_count
HR	2
Finance	2

Question 4

SELECT student_id,
name,
age,
department

FROM students

WHERE age **BETWEEN** 21 **AND** 23;

student-id	name	age	department
2	Bob	22	HR
4	Diana	23	Finance
5	Eve	22	HR
3	Charlie	21	IT

Question 5

SELECT student_id,
name,
age,
department

FROM students

WHERE department =('IT', 'HR')

AND age > 21;

student-id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

Question 6

SELECT department,
SUM(credits) **AS** total_credits;
FROM courses

GROUP BY department

HAVING SUM(credits) >5;

department	total_credits
IT	11

Question 7

SELECT course_id,
credits,
department,
course_name

FROM courses

WHERE credits<>4;

course_id	Course_name	department	credits
101	SQL basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

Question 8

SELECT course_id,
course_name,
credits

FROM courses

ORDER BY credits **DESC**

LIMIT 3;

course_id	Course_name	credits
102	Python	4
103	Data science	4
101	SQL	3

Question 9

```
SELECT MAX(grade) AS max_grade,  
       MIN(grade) AS min_grade,  
       AVE(grade) AS avg_grade  
FROM enrollment;
```

max_grade	min_grade	avg_grade
90	78	84.6

Question 10

```
SELECT course_id,  
       COUNT (*) AS enrollment_count,  
FROM enrollments  
GROUP BY course;
```

Course_id	Enrollment_count
101	1
102	1
103	1
104	1
105	1

Question 11

```
SELECT department,  
       SUM(salary) AS total_salary,  
       SUM(bonus) AS total_bonus  
FROM salaries  
GROUP BY department;
```

department	total_salary	total_bonus
IT	12200	10500
HR	10900	7500
Finance	70000	6000

Question 12

```
SELECT department,  
       AVG (salary) AS avg_salary  
FROM salaries  
GROUP BY department,  
HAVING AVG(salary)>55000;
```

Department	AVG_SALARY
IT	61000
Finance	70000

Question 13

```
SELECT employee_id,  
       name,  
       salary,  
       bonus,  
       SUM(salary) AS total_salary,  
       SUM(bonus) AS total_bonus,  
       SUM(salary+bonus) AS total-compensation  
FROM salaries  
HAVING SUM(salary+ bonus)>60000
```

Employee_id	Name	Salary	Bonus	Total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	6200	5500	67500

Question 14

```
SELECT department,  
       SUM(budget) AS total_budget,  
       AVG(budget)AS avg_budget
```

FROM projects

GROUP BY department,

HAVING AVG(budget)> 70000;

department	total_budget	avg_budget
IT	270000	135000
Finance	80000	80000

Question 15

SELECT project_id,

project_name,

department,

budget

FROM projects

WHERE budget **BETWEEN** 50000 **AND** 120000

AND department!='Marketing';

project_id	project_name	department	budget
1	AI App	IT	120000
2	Payroll system	Finance	80000
5	HR Portal	HR	50000